Elmergib University

Faculty of Information Technology

Midterm Exam 2024-2025 Exam Duration: 1 hr

Name: …….

Student id:……

Question 1: What is Python ? **(2 marks)**

Question 2: What are the uses of the Python language? **(2 marks)**

Question 3: What are the features of the Python language? **(2 marks)**

Question 4: (14 marks)

In front of you is a table with codes in the Python language. Write what is the output you get in the empty column.

Output	Python code	

[1]
[1, 2]
[4, 3]
	`def func(x, lst=[]):`
	`  lst.append(x)`
	`  return lst`
	
	`print(func(1))`
	`print(func(2))`
	`print(func(3, [4]))`
		1 mark

6
7
	`a = 5`
	`a = 7`
	`def func():`
	`  a = 3`
	`  print(a + a)`
	`func()`
	`print(a)`
		1 mark

10	`x = 10`
	`def func():`
	`  return x`
	`print(func())`
		1 mark

_Small_Small	`def foo(x):`
	`  if x > 5:`
	`  return "Big"`
	`  else:`
	`  return "_Small"`
	`result = foo(5) + foo(2)`
	`print(result)`
		1 marks

10
7
	`x = 7`
	`def func(x):`
	`  x += 3`
	`  return x`
	`print(func(x))`
	`print(x)`
		1 marks

2.0	`def foo(x):`
	`  if x == 0:`
	`  return "Error"`
	`  else:`
	`  return x / 3`
	
	`print(foo(foo(18)))`
		1 marks

[1,2,3,[1,2,3]]	`a = [1, 2, 3]`
	`b = [1, 2, 3]`
	`a.append(b)`
	`print(a)`
		2 marks

Sum: 8
Product: 15
	`def outer_func(a, b):`
	`  def inner_sum(x, y):`
	`  return x + y`
	
	`  def inner_product(x, y):`
	`  return x * y`
	
	`  sum_result = inner_sum(a, b)`
	`  product_result = inner_product(a, b)`
	
	`  return sum_result, product_result`
	
	`result = outer_func(5, 3)`
	`print("Sum:", result[0])`
	`print("Product:", result[1])`
		2 marks

84	`def add(a, b):`
	`  return a + b`
	`def subtract(a, b):`
	`  return a - b`
	`result = add(10, subtract(77, 3))`
	`print(result)`
		2 marks

	`def func(lst):`
	`  lst.append(4)`
	`my_list = [1, 2, 3]`
	`func(my_list)`
	`print(my_list)`
		2 marks

Question 5 : (10 marks)

In the code below the `search_contact` function is missing a crucial part. Complete the missing part in the function to enable the search functionality. The function should search for a contact by name and print the contact's details if found. If no contact is found, it should print a message indicating that no contact was found with the given name.

```python
for contact in contacts:
        if contact['name']
== search_name:
            print(f"Name:
{contact['name']}, Phone:
{contact['phone']}, Email:
{contact['email']}")
            found = True
            break
```

```python
contacts = []

def add_contact():
    name = input("Enter name: ")
    phone = input("Enter phone number: ")
    email = input("Enter email address: ")
    contact = {'name': name, 'phone': phone, 'email':
email}
    contacts.append(contact)

def view_contacts():
    if not contacts:
        print("No contacts available.")
    else:
        for contact in contacts:
             print(f"Name: {contact['name']}, Phone:
{contact['phone']}, Email: {contact['email']}")

def search_contact():
    search_name = input("Enter the name to search: ")
    found = False
    # Missing part starts here
    # for contact in contacts:
    #     if contact['name'] == search_name:
    #         print(f"Name: {contact['name']}, Phone:
{contact['phone']}, Email: {contact['email']}")
    #         found = True
    #         break
    # Missing part ends here
    if not found:
        print(f"No contact found with the name
{search_name}.")

def main():
    while True:
        print("\nOptions:")
        print("1. Add Contact")
        print("2. View Contacts")
        print("3. Search Contact")
        print("4. Exit")
        choice = input("Choose an option: ")
        if choice == '1':
            add_contact()
        elif choice == '2':
            view_contacts()
        elif choice == '3':
            search_contact()
        elif choice == '4':
            print("Goodbye!")
            break
        else:
            print("Invalid choice. Please try again.")
main()
```

Good Luck !